House Price Prediction with Frozen Visual Features and Kernel Fusion

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I. Introduction

Estimating a house's sale price involves size, condition, location and market conditions. Exterior photographs reveal some of this information: computer vision can describe architectural form, surface appearance, landscaping and surrounding scenes. However, a single view provides incomplete coverage and does not directly reveal floor area, an exact address or transaction conditions. We therefore investigate how to extract more informative visual features from house photographs to predict sale prices more accurately.

Handcrafted descriptors summarize local gradients and textures; AlexNet [1] learns hierarchical features. CLIP [2], SigLIP2 [3] and Perception Encoder [4] enable transfer through frozen visual features and separately fitted regression.

Our study follows three stages, each testing a limitation exposed by the preceding experiments. SIFT with a visual vocabulary first tests whether local appearance predicts price. An AlexNet-style regressor then undergoes controlled training, augmentation and architecture experiments. Its remaining error motivates frozen pretrained features, for which we investigate image coverage, feature transformation and regression before combining complementary branches.

The selected fusion achieves validation MSE 52,201.21 and MAE 159.58, with public Kaggle MSE 47,291.35 after full-data refitting. We examine its gains and persistent high-price underestimation.

II. Methods

A. Data, Metrics and Evaluation Protocol

The supplied Southern California dataset contains 8,000 labelled training images and 3,000 unlabelled test images; prices are in thousand USD. General-purpose pretrained encoders provide transferable features.

We split the data into 6,399 training and 1,601 validation images with seed 2026, keeping identified duplicate or related-image groups together. For N evaluated images with true prices y_i and predictions $\hat{y}_i$, we report

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2, \qquad \mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|. \quad (1)$$

MSE, the primary metric, emphasizes large errors; MAE measures average absolute error. Their units are (thousand USD)^2 and thousand USD, respectively. Signed residuals $\hat{y}_i - y_i$ describe bias.

Stochastic models are screened with seed 2026; promising candidates are repeated with seeds 2027 and 2028. Adoption requires lower mean MSE and at least two seed-matched wins; we report three-seed means and paired outcomes. Frozen-feature regressors use grouped three-fold training out-of-fold (OOF) selection, fitting codebooks, scalers, PCA and supervised transforms within each fold.

B. Baselines and Controlled CNN Development

Beyond a training-mean reference, we extract grayscale SIFT descriptors [5], [6] and build a 256-word visual vocabulary [7]. Histograms undergo L1–square-root–L2 normalization and standardization before Ridge regression [8] ($\alpha = 10$). Histogram pooling discards spatial arrangement, motivating learned CNN features.

Our AlexNet-style regressor [1] has five convolutional layers and a 9216–4096–4096–1 head, initialized from scratch. After preprocessing comparisons, the selected input is a direct 224×224 resize, RGB normalization with mean and SD 0.5, and no filtering. The retained configuration uses AdamW [9] (10^{-4} learning rate and weight decay), dropout [10], [11] 0.5, and effective batch 32 through microbatch 4 and eight-step accumulation. Training uses prices divided by 1,000; predictions are restored to dataset units. Runs last at most 60 epochs, with a minimum of 40 and patience 15 under a 0.1% improvement stopping rule; the checkpoint with strictly lowest validation MSE is retained.

Optimization screens use learning rates 3×10^{-5}, 10^{-4}, 3×10^{-4} and 10^{-3}, effective batches 16/32/64 and dropout 0/0.2/0.5. We compare Adam [12], cosine decay [13] without restarts, effective batch size, regularization and initialization [14], [15], then test augmentation. Geometric and degradation transforms are evaluated separately; horizontal flip uses probability 0.5. Architecture experiments test stacked 3×3 convolutions [16] replacing the second 5×5 layer, an Inception-style multi-scale block [17], and a projected residual connection [18]. To reduce dense-head capacity, we project the 6×6 convolutional grid to 128 channels, add positional information and normalization, and average the 36 tokens before linear regression. We then test a one-layer Transformer [19] with dimension 128, four heads and a 512-dimensional feed-forward layer. Each change is evaluated against its corresponding control.

In a separate seed-2026 screen, brightness and contrast are each multiplied by a factor sampled uniformly from

$[0.9, 1.1]$ on every training image before resizing and normalization. Both are compared with a freshly trained unaugmented AlexNet using the same initialization, split and training settings.

C. Frozen Representations, Views and Regression

Frozen CLIP ViT-B/32 with standardized Ridge establishes the transfer reference; final branches use SigLIP2 So400m NaFlex, PE-Core-L/14 and PE-Core-G/14. NaFlex preserves native aspect ratio with 576 patches and produces 1,152-dimensional features. Because a single crop can remove useful context, PE-L combines three 1,024-dimensional views: the official input, a 336-pixel gray-padded whole frame, and a central crop spanning 60% of image width and height. PE-G combines its official 448-pixel view with a gray-padded whole frame. Features are individually unit-normalized, concatenated and normalized again, yielding 3,072 PE-L and 2,560 PE-G dimensions.

RBF kernel ridge regression [20] models nonlinear feature–price relationships. For unit features, squared distance equals twice one minus cosine similarity; γ controls locality. For x,

$$K_{ij} = \exp(-\gamma \|x_i - x_j\|^2), \quad c = (K + \alpha I)^{-1}(y - \bar{y}\mathbf{1}), \tag{2}$$

with prediction $\bar{y} + k(x)^T c$ and non-negative clipping. Ordinary kernel searches use $\gamma \in \{1, 4, 16\}$ and $\alpha \in \{0.01, 0.1, 1, 10\}$ on training folds. NaFlex selects $\gamma = 1$, $\alpha = 0.1$.

PE-L adapts generic similarity using fold-fitted PLS [21], [22]. Of sixteen fitted components, the first four standardized scores s enter the product kernel

$$k(x, x') = \exp\left(-\|x - x'\|^2 - 0.1\frac{\|s - s'\|^2}{4}\right), \tag{3}$$

with $\alpha = 1$. The scores introduce price supervision while the encoder remains frozen.

In contrast to supervised PLS, PCA [23] compresses features along their principal variation directions without using prices. For PE-G, we test 64, 256 and 1,024 dimensions with whitening powers 0 and 0.5, followed by unit normalization. Since this changes distances, we also recalibrate the original $\gamma = 4$:

$$\gamma' = 4\frac{\text{median}\,\|x_i - x_j\|^2}{\text{median}\,\|z_i - z_j\|^2}, \tag{4}$$

using 8,192 fixed non-self pairs drawn only from fitting-fold images. Here z is the transformed unit vector. The selected transform is non-whitened PCA256 with $\alpha = 1$.

D. Fusion and Final Refit

Following stacked regression [24], we combine branch predictions using non-negative weights, additionally constrained to sum to one. Weights are fitted to training OOF predictions and evaluated by meta cross-fitting: each fold is predicted using weights fitted on the remaining folds. A new branch is retained when meta-CV MSE improves by at least 1%, at least two of three folds improve, and validation MSE also decreases.

TABLE I
REPRESENTATIVE VALIDATION RESULTS.

Method	MSE	MAE
Reference baselines		
Training mean	151,840	294.47
SIFT–BoVW–Ridge	145,337	288.66
A. Representation route		
AlexNet	127,774	264.52
AlexNet + flip	126,810	268.02
Flip AlexNet + mean head	122,334	261.48
CLIP B/32 + Ridge	91,821	223.27
B. Frozen branches and fusion		
NaFlex + RBF	63,086	181.28
PE-L (3 views) + RBF	59,213	172.90
PE-L (3 views) + PLS/RBF	59,483	171.73
PE-G (official view) + RBF	55,097	165.80
PE-G (2 views) + RBF	53,253	161.88
PE-G (2 views) + PCA/RBF	52,806	160.67
Three-encoder fusion	**52,201**	**159.58**

$n = 1{,}601$; MSE: (thousand USD)^2; MAE: thousand USD. SIFT and the three AlexNet rows are three-seed means; frozen-feature rows use selected fits.

The final weights are NaFlex 0.108065, PE-L PLS 0.080806 and PE-G PCA 0.811129. For the Kaggle submission, we retain the selected hyperparameters and weights and refit the transforms and regression heads on all 8,000 labelled images, recomputing PCA and bandwidth statistics from this training set. The fitted model then predicts the 3,000 test prices.

III. RESULTS

A. Local Evidence and the Need for Learned Representations

SIFT–BoVW–Ridge reduces training-mean MSE 151,839.69 to $145,336.73 \pm 1,091.58$, revealing limited local-appearance signal. AlexNet improves this to 128,415.68 with padding and 127,774.42 with direct resizing (Table I), motivating training and capacity experiments.

B. Testing Training and Capacity Before Transfer

Seed-matched trials provide little support for optimization changes. Adam at learning rate 0.001 improves seed 2026 but loses the other two comparisons, increasing mean MSE by 509.89; cosine decay, alternative batches and lower dropout also fail adoption. Horizontal flip instead reduces mean MSE by 964.41 with two paired wins, although MAE rises from 264.52 to 268.02. Erasing improves the initial screen but wins only one of three paired reruns and is rejected. Thus, adoption follows repeated evidence rather than the best screening score.

To test whether the large prediction head limits learning, we replace it with mean pooling. All three seeds improve, reducing MSE by 4,475.78 relative to flip AlexNet. Small-kernel, multi-scale and residual screens fail their controls; the Transformer increases single-seed MSE by 3,615.11 over the compact head. Greater capacity therefore does not resolve this setting's prediction difficulty.

Photometric changes reveal a further trade-off. Against a fresh seed-2026 control (MSE 131,389.94, MAE 282.64), brightness gives 131,433.91 and 274.50; contrast gives 128,860.53 and 269.01. Contrast improves overall MSE by

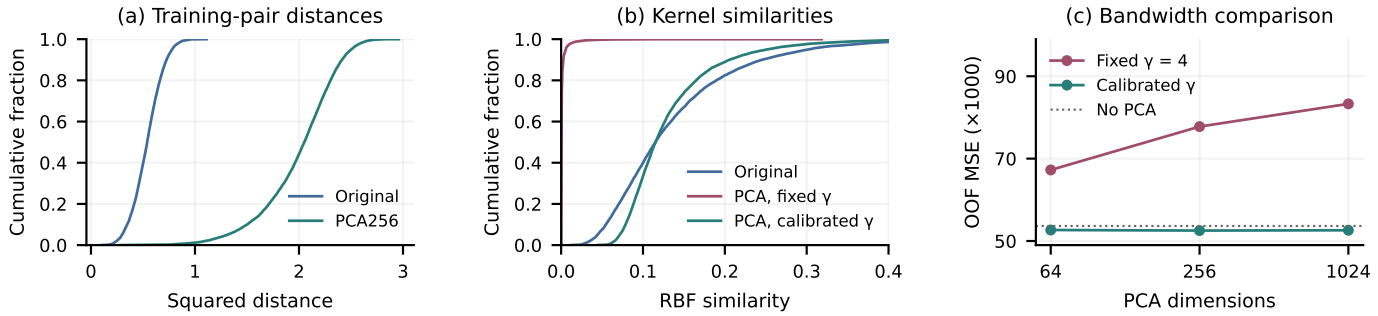


Fig. 1. PE-G geometry and regression. (a,b) Empirical CDFs of distances and kernel similarities for 8,192 training pairs; (c) grouped three-fold OOF MSE for non-whitened PCA. Transforms and bandwidths are fitted within each fold. Panel (b) shows the similarity range 0–0.4.

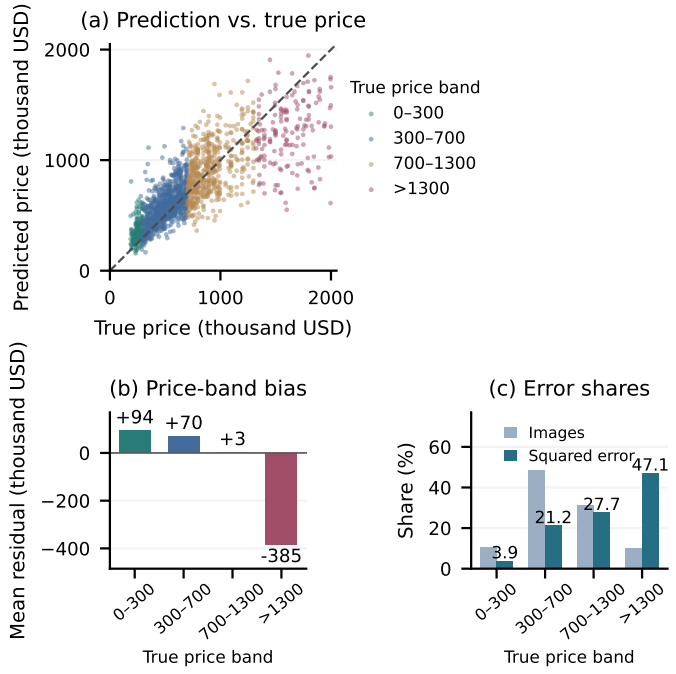


Fig. 2. Selected fusion on 1,601 validation images. (a) Predicted versus true price; the dashed line is perfect prediction and colors follow the price bands in (b). (b) Mean residual $\hat{y} - y$. (c) Image and squared-error shares. Bands are $(0, 300]$, $(300, 700]$, $(700, 1300]$ and above 1,300 thousand USD.

1.93% but worsens the above-1,300 tail by 17.14%; brightness worsens it by 3.86%. Thus, lower average error need not help expensive properties. Both remain single-seed screens.

CLIP with Ridge reaches MSE 91,820.73, motivating richer transferred features. Both representation and readout change in this route comparison.

C. Recovering Context and Repairing Kernel Geometry

Supplementing PE-G's official view with the whole frame reduces validation MSE from 55,097.49 to 53,252.81 and OOF MSE from 55,679.73 to 53,646.46, supporting broader coverage.

However, compressing these features with PCA256 and fixed $\gamma = 4$ worsens OOF MSE to 77,771.43. The training median squared distance changes from 0.5402 to 2.0515, so we recalibrate γ to 1.05323 (Fig. 1). OOF MSE falls to 52,531.27; validation MSE reaches 52,805.77 and MAE 160.67. The distance shift explains why compression must be evaluated together with the downstream similarity scale; it does not establish that PCA removes only irrelevant information.

Supervised adaptation is also not uniformly beneficial: PE-L PLS slightly worsens validation MSE relative to its three-view RBF control, while improving MAE. Its usefulness must consequently be judged through complementarity, rather than assumed from price supervision.

TABLE II
FUSION ABLATION AND STRONGEST-BRANCH COMPARISON.

Configuration	Val. MSE	Meta MSE
PE-G PCA only	52,805.77	52,531.27
No NaFlex	52,597.50	52,359.30
No PE-L PLS	52,194.81	52,244.24
No PE-G PCA	57,372.90	57,694.20
Equal weights	53,847.68	54,214.33
Full fusion	52,201.21	52,210.22

After branch removal, remaining weights are refitted on training OOF predictions only. Validation n=1,601; meta uses three-fold cross-fitting on training OOF predictions.

D. Complementarity and Remaining Valuation Ambiguity

Adding PE-G PCA lowers meta-CV MSE from 53,052.57 to 52,210.22 with all three folds improving, and validation MSE from 52,524.78 to 52,201.21. Public Kaggle MSE changes from 47,837.69 to 47,291.35.

Table II tests whether fusion adds value beyond its strongest branch. It improves on PE-G PCA alone by 1.14%; removing PE-G has the largest cost, whereas equal weights dilute its contribution. Removing PE-L slightly improves validation MSE but worsens meta-CV, so its benefit is marginal and criterion-dependent. These are branch-removal tests, not isolated tests of PCA or PLS.

Aggregate success leaves a new problem: mean residual $+6.51$ masks low-price overestimation ($+93.72$ at prices up to 300) and high-price underestimation (-385.16 above 1,300). The 159 highest-price images contribute 47.13% of squared error (Fig. 2); the 161 largest errors contribute 63.30%. RMSE is 228.48 and median absolute error 113.41.

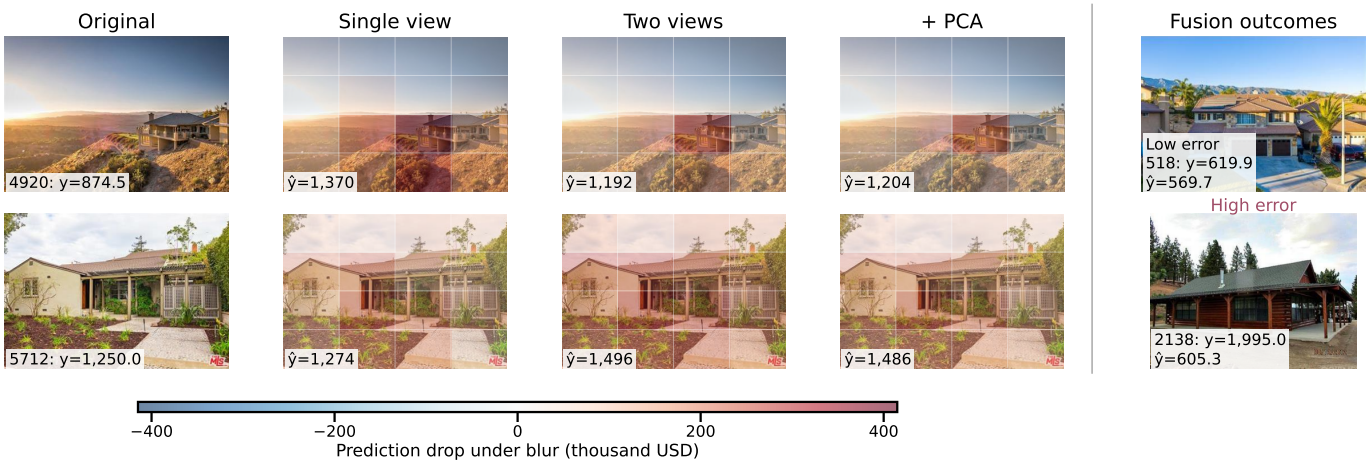


Fig. 3. Left four columns: PE-G regional sensitivity under 4×4 blur replacement; red indicates a prediction decrease, blue an increase, on one shared scale. Cases 4920/5712 are the largest improvement/worsening after adding the whole frame among 1,543 landscape validation images. Right column: separate final-fusion cases, 518 at the 25th absolute-error percentile and retained failure 2138. Prices are in thousand USD.

E. Qualitative Cases and Regional Sensitivity

Figure 3 contrasts fusion case 518 (absolute error 50.17) with severely underestimated case 2138. Both show complete buildings, so adequate coverage alone does not ensure accuracy.

Regional perturbation [25] tests a narrower question. Replacing 4×4 regions with gray pixels or radius-8 blur gives the largest responses near house regions (Fig. 3). However, the two-view gray/blur correlation for 5712 is only 0.267. These maps describe perturbation sensitivity, not reliable identification of price-related attributes.

IV. Discussion

The three routes reveal complementary requirements for visual valuation. Handcrafted SIFT features explicitly represent local appearance, but visual-word pooling discards spatial relationships, potentially limiting discrimination between properties with different building and environmental layouts. This highlights the importance of organizing image evidence.

The scratch-CNN route makes this organization learnable, but requires learning features and price relationships from limited supervision. The compact head suggests that matching capacity to available labels can be more useful than architectural complexity. Greater expressive power therefore does not automatically produce more informative price features.

Frozen pretrained encoders supply knowledge learned beyond the price-labelled dataset, allowing task learning to focus on regression. Adaptation remains essential: views determine which evidence survives preprocessing, while transforms and kernel geometry connect visual similarity to price. Caching enables reuse on limited hardware, although dense kernels constrain scalability; fusion exploits complementary representations with unequal contributions.

Together, accurate prediction requires informative representations, adequate coverage and a mapping aligned with valuation. Appearance provides partial evidence of value; unobserved location, floor area and transaction conditions may explain remaining ambiguity. Next, we would compare errors by building coverage, occlusion and scene context before testing targeted views. High-price sampling should be checked for error shifts in ordinary-price groups, while contrast augmentation needs multi-seed confirmation. Reused development data and prior branch selection also warrant an independent grouped evaluation; meta cross-fitting alone does not remove this selection bias.

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